SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID – 26167**
- **Problem Statement Title- SatQuery AI: An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries**
- **Theme- Space Technology**
- **PS Category- Software**
- **Team ID- (as registered on portal)**
- **Team Name (Registered on portal)- Lunar Circle**

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SatQuery AI

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

What it is

An agentic vision-language assistant — upload GeoTIFF satellite imagery, ask in plain language, get an evidence-grounded answer with masks/boxes, confidence and an auditable execution trace.

Covers the full prescribed scope

Single optical/SAR image · co-registered optical–SAR pair · bi-temporal pair — across VQA, captioning, grounding, change-VQA and optical–SAR fusion.

How it addresses the problem

Replaces fragmented single-task RS tools: a controller validates every input and routes each query to fine-tuned specialist models — no GIS expertise needed.

Innovation & uniqueness

Agentic orchestration (not one generic VLM) · LoRA adaptation on BigEarthNet.txt (464,044 co-registered S1 SAR + S2 pairs, 9.6M annotations) · rejects incompatible inputs · every answer ships visual evidence.

Working prototype — mission console (all 5 tasks, 3 input scopes)

LIVE NOW — TRY IT

gangadhar017.github.io/Lunar_circle

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TECHNICAL APPROACH

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Technologies to be used

Python · PyTorch · HF Transformers · Qwen2.5-VL + LoRA (4-bit) · LangGraph · GDAL·rasterio · FastAPI · Docker · Pages + Render

Methodology & process — agentic pipeline (working prototype)

1 · Query + Imagery

NL question + GeoTIFF/TIFF (optical/MS, SAR, pairs)

2 · Input Validator

count · modality · format · CRS · co-registration

3 · Agentic Controller

classifies task, plans workflow (LangGraph)

4 · Task Router

selects from registry; permitted params only

SPECIALIST MODEL REGISTRY — remote-sensing fine-tuned (BigEarthNet.txt · LoRA)

RS-VQA

single-image VQA (mandatory baseline)

Caption + Grounding

scene description · region localisation

Change-VQA

bi-temporal change description & maps

Optical–SAR Fusion

joint built-up / water extraction

5 · Output Integrator

fuses text + spatial outputs · estimates confidence · attaches evidence

6 · Evidence-Grounded Response

answer + overlays/boxes + confidence + auditable summary + report

SCAN — LIVE PIPELINE DEMO

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FEASIBILITY AND VIABILITY

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Analysis of feasibility

- **100% open data**
  BigEarthNet.txt, VRSBench, RSVQA, CDVQA — training & evaluation start on day one.
- **Strong open baselines**
  GeoChat, RemoteCLIP and change models exist — we adapt, not reinvent.
- **Fits a free GPU**
  LoRA fine-tunes <1% of parameters — trains on a single Colab/Kaggle T4.
- **Modular teamwork**
  Each specialist is built & tested independently, then registered.
- **Already deployed**
  Frontend live on GitHub Pages; Dockerised FastAPI backend one click from Render.

Potential challenges & risks	Strategies to overcome
Limited GPU compute for VLM training	LoRA/PEFT + 4-bit quantised inference; frozen backbone, small adapters.
GeoTIFF & multi-band sensor handling	GDAL preprocessing: band selection, normalisation, tiling, CRS checks.
Domain gap — Cartosat-2S / RISAT eval	Multisensor S1+S2 training + SAR-style speckle & resolution augmentation.
VLM hallucination on unfamiliar scenes	Grounded masks/boxes, confidence thresholds, strict input validator.
Live orchestration failure	Deterministic routing + full trace + graceful fallback to single-image VQA.

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IMPACT AND BENEFITS

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Potential impact on the target audience — no GIS training required

Agriculture
crop & land-cover monitoring each season by simple queries

Disaster response
SAR sees through clouds — flood answers day or night

Urban planning
“has built-up area grown?” answered with change maps

Forests & ecology
deforestation & vegetation change tracked over time

Water resources
water bodies grounded across optical + SAR

Infrastructure
assets mapped & monitored from multimodal imagery

Benefits of the solution

SOCIAL	Democratises ISRO Earth-observation data — district officials, farmers' bodies and NGOs question imagery in plain language.
ECONOMIC	Hours of manual multi-tool GIS analysis → seconds per query; one platform replaces many single-task systems.
ENVIRONMENTAL	Continuous monitoring of forests, water and sprawl; SAR works through clouds, day & night — critical in floods and cyclones.
STRATEGIC	Multiplies the value of Cartosat & RISAT missions; auditable, evidence-grounded AI fit for official governance use.

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RESEARCH AND REFERENCES

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Datasets (as prescribed by PS 26167)

BigEarthNet.txt — training

arXiv:2603.29630 · txt.bigearth.net — 464,044 co-registered Sentinel-1 SAR + Sentinel-2 pairs, 9.6M annotations (captions · VQA · referring-expression detection)

VRSBench — evaluation

github.com/lx709/VRSBench — captioning, grounding & VQA

RSVQA — evaluation

rsvqa.sylvainlobry.com — remote-sensing VQA (LR/HR)

CDVQA — evaluation

github.com/YZHJessica/CDVQA — change-based VQA

Key research we build on

GeoChat (CVPR 2024)

grounded VLM for remote sensing — adaptation base

RemoteCLIP (2024)

RS vision-language representation learning

RSVQA (IEEE TGRS 2020) · CDVQA (IEEE TGRS 2022)

task formulations for RS-VQA and change-VQA

LoRA (ICLR 2022)

parameter-efficient fine-tuning under low compute

Evaluation-ready: prescribed public test splits with normalised scores (accuracy · BLEU/CIDEr · IoU); multisensor training targets the ISRO/SAC Cartosat-2S + RISAT hidden set.

LIVE PROTOTYPE gangadhar017.github.io/Lunar_circle SOURCE + BACKEND github.com/Gangadhar017/Lunar_circle